Prediction of Concrete Compressive Strength Using Concrete Mixture Characteristics

Mohamed Alsaleh
Department of Civil and Environmental Engineering
University of Illinois Urbana-Champaign
Urbana, IL, USA
alsaleh4@illinois.edu

Jingchun Wang
Department of Civil and Environmental Engineering
University of Illinois Urbana-Champaign
Urbana, IL, USA
jw198@illinois.edu

Khwanjira Phumphid
Department of Civil and Environmental Engineering
University of Illinois Urbana-Champaign
Urbana, IL, USA
kp70@illinois.edu

Rae Zhu
Department of Civil and Environmental Engineering
University of Illinois Urbana-Champaign
Urbana, IL, USA
rzhu15@illinois.edu

***Abstract*—This project aims to investigate the relationships between concrete mixture characteristics, age, and compressive strength using exploratory data analysis and predictive modeling. Multiple statistical and machine-learning approaches will be evaluated to estimate concrete compressive strength and identify influential mixture characteristics. The findings are expected to provide insights into concrete strength development and inform future mixture design adjustments.**

***Index Terms*—Concrete Compressive Strength, Machine Learning, Regression, Feature Importance**

I. Dataset Description

A. Dataset Source and Overview

This project will use the Concrete Compressive Strength dataset available through the UCI Machine Learning Repository [1]. The dataset was originally provided by Prof. I-Cheng Yeh of Chung-Hua University. The data were developed from laboratory measurements of concrete mixtures, where the actual compressive strength of each mixture was determined at a specified age. The dataset is provided in raw, unscaled form and is available as an Excel file (Concrete_Data.xls).

Dataset source: UCI Machine Learning Repository, *Concrete Compressive Strength* dataset

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The dataset contains 1,030 observations and nine quantitative variables. Eight variables describe the concrete mixture composition and age and are used as input variables. The ninth variable, concrete compressive strength, is the output variable. The dataset contains no missing values.

B. Dataset Variables

The variables included in the dataset are summarized in Table Table I.

TABLE I
Description of dataset variables.

Variable	Unit	Description	Role
Cement	kg/m^3	Amount of cement in the concrete mixture	Input
Blast Furnace Slag	kg/m^3	Amount of blast furnace slag in the mixture	Input
Fly Ash	kg/m^3	Amount of fly ash in the mixture	Input
Water	kg/m^3	Amount of water in the mixture	Input
Superplasticizer	kg/m^3	Amount of superplasticizer in the mixture	Input
Coarse Aggregate	kg/m^3	Amount of coarse aggregate in the mixture	Input
Fine Aggregate	kg/m^3	Amount of fine aggregate in the mixture	Input
Age	days	Age of the concrete specimen at testing	Input
Concrete Compressive Strength	MPa	Measured compressive strength of the concrete	Output

Therefore, the primary response variable for this project will be **concrete compressive strength in MPa**, while the remaining eight variables will be considered potential predictors.

C. Exploratory Data Analysis

Fig. 1 shows the distribution of concrete compressive strength across different specimen ages. Overall, the median value of compressive strength increases with age, especially during the early stage. For example, the median compressive strength increases from approximately 16 MPa at 3 days to 34 MPa at 28 days, and continue to increase to 52 MPa at 56 days. However, the compressive strength varies within each age group. This variation existing within each age group suggests that curing age alone cannot fully explain the differences in compressive strength.

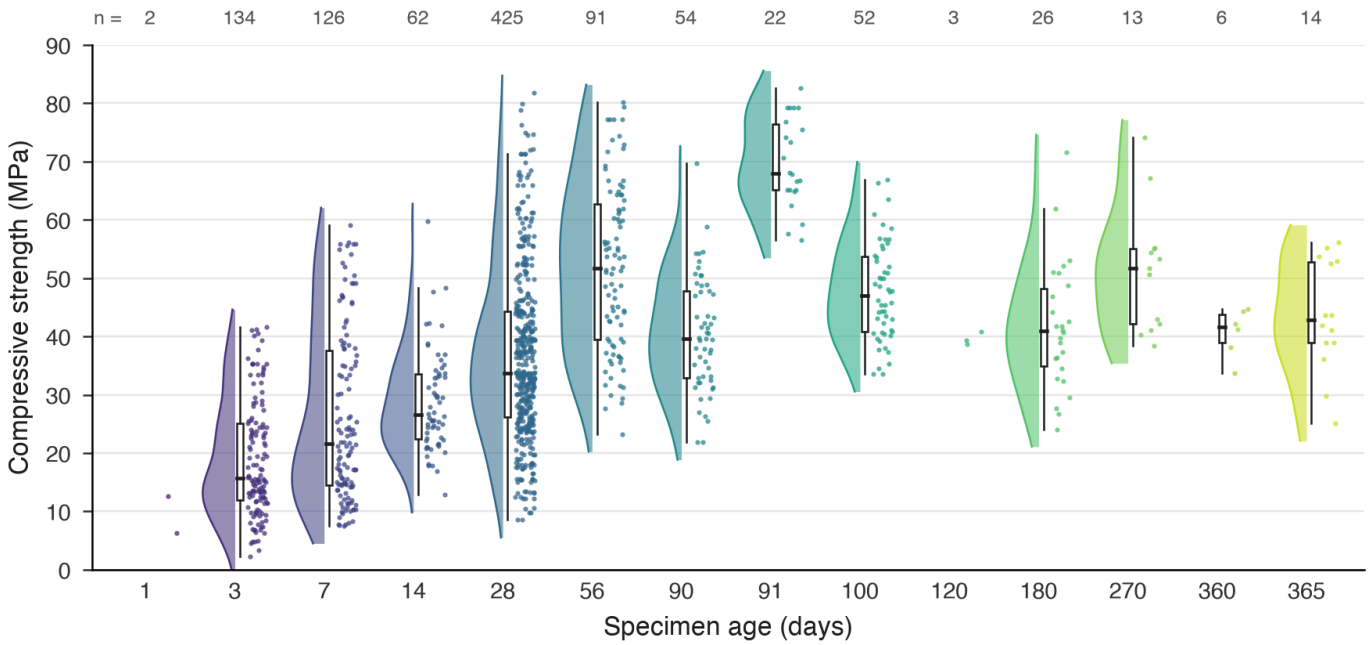


Fig. 1. Distribution of compressive strength by specimen age.

In addition, the strength distributions across the different age groups do not necessarily show a consistent upward trend. For example, the 91-day group has a much higher median strength than the 90-day group. This difference does not necessarily indicate a sudden rise in strength, because the specimens tested at these ages may have different mixture compositions. The sample sizes are not consistent either: 425 specimens were tested at 28 days, while only 22 were tested at 91 days. Therefore, comparisons across age groups may not be entirely valid, and the effects of mixture composition should also be considered.

Fig. 2 presents the pairwise relationships and Spearman's rank correlation coefficients among the selected variables. Specimen age has the strongest positive correlation with compressive strength (ρ=0.60), followed by cement content (ρ=0.48) and high-performance water-reducing agent(superplasticizer) content (ρ=0.35). In contrast, water content showed a negative correlation with compressive strength (ρ=-0.31), suggesting that mixtures with higher water content tend to have lower strength.

The relationships among the mixture components are also important. For example, fly ash and superplasticizer show a positive correlation (ρ = 0.45), while water and superplasticizer show a negative correlation (ρ = −0.69). These relationships indicate that some predictors are associated with each other, which should be considered when developing predictive models. Furthermore, the relationship between age and strength is described more accurately by the Spearman correlation coefficient (ρ = 0.60) than by the Pearson correlation coefficient (r = 0.33), indicating that this relationship may be monotonic but nonlinear.

Fig. 3 illustrates the relationship between binder composition and 28-day concrete compressive strength using a ternary diagram of cement, blast furnace slag, and fly ash proportions. Comparing specimens at the same testing age allows the relationship between binder composition and compressive strength to be examined without differences in specimen age affecting the comparison.

The figure shows that some mixtures with higher cement proportions achieve relatively high compressive strength. However, high strength is also observed in mixtures containing slag or fly ash, suggesting that partially replacing cement with these materials does not necessarily result in lower strength. Considerable variation in compressive strength is observed even among mixtures with similar binder proportions. This indicates that binder composition alone cannot fully explain the differences in strength. Other factors, such as water content, total binder content, and chemical admixtures, may also contribute to the observed variation. Therefore, multiple mixture characteristics should be considered when developing predictive models for concrete compressive strength.

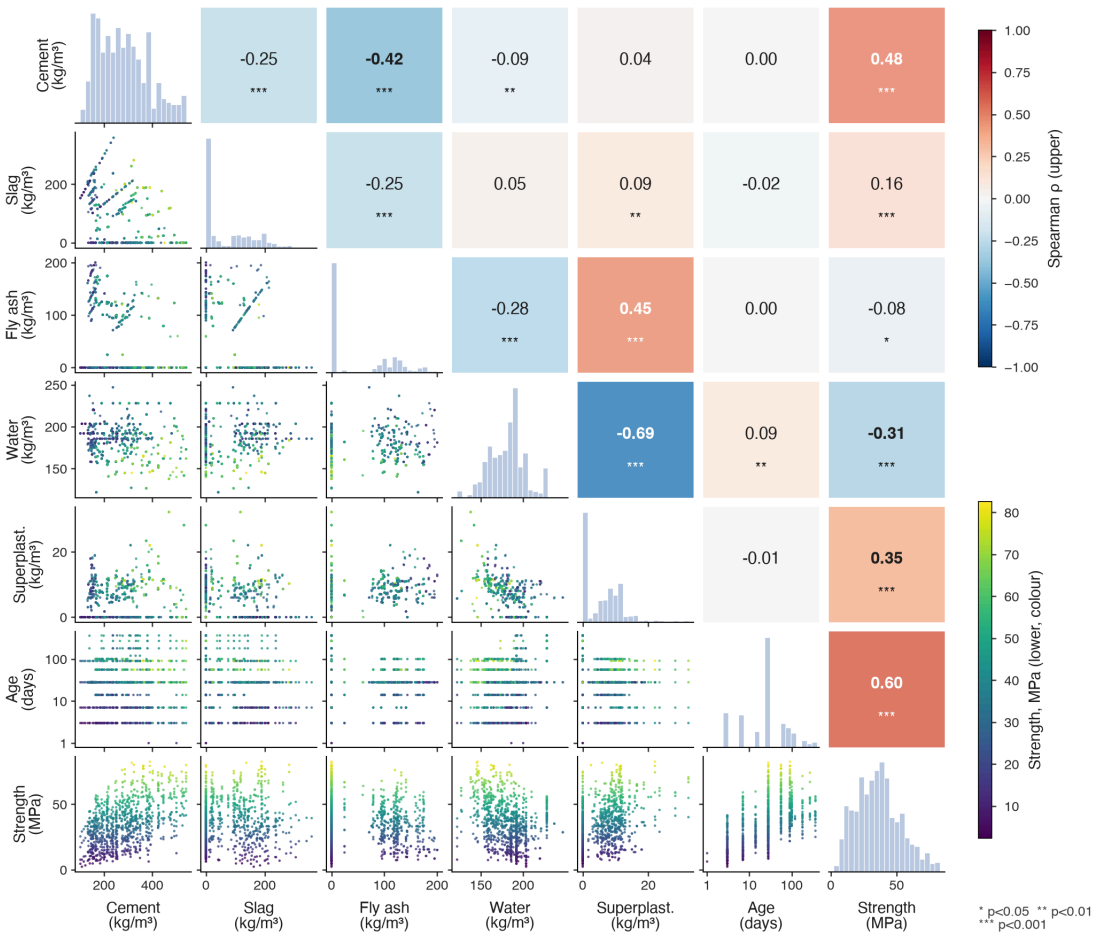


Fig. 2. Pairwise relationships and Spearman rank correlations among mixture variables, age, and compressive strength.

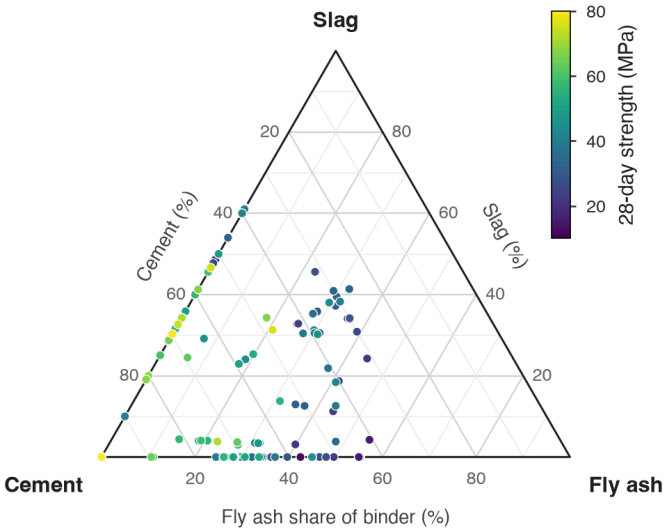


Fig. 3. Binder composition and 28-day compressive strength.

Fig. 4 shows the relationship between the water-to-binder ratio and the concrete compressive strength, with specimen age represented by color. A clear decreasing trend can be observed, as mixtures with lower water-to-binder ratios typically achieve higher compressive strengths. For example, specimens with water-to-binder ratios below approximately 0.35 exhibit strengths ranging from about 20 to over 80 MPa, while most specimens with ratios above 0.70 have strengths below 30 MPa. This relationship is consistent with the role of water in cement-based materials. This pattern is consistent with the understanding that excessive water relative to binder content can increase porosity and reduce concrete strength.

However, differences still exist among specimens with similar water-to-binder ratios. The color distribution indicates that specimens tested at later ages generally show achieve strength than those tested at earlier ages with comparable water-to-binder ratios. This indicates that both specimen age and water-to-binder ratio are associated with compressive strength. Other mixture characteristics may also explain some of the remaining variation. Therefore, the relationship should be interpreted as an association rather than evidence of a causal effect from this dataset alone.

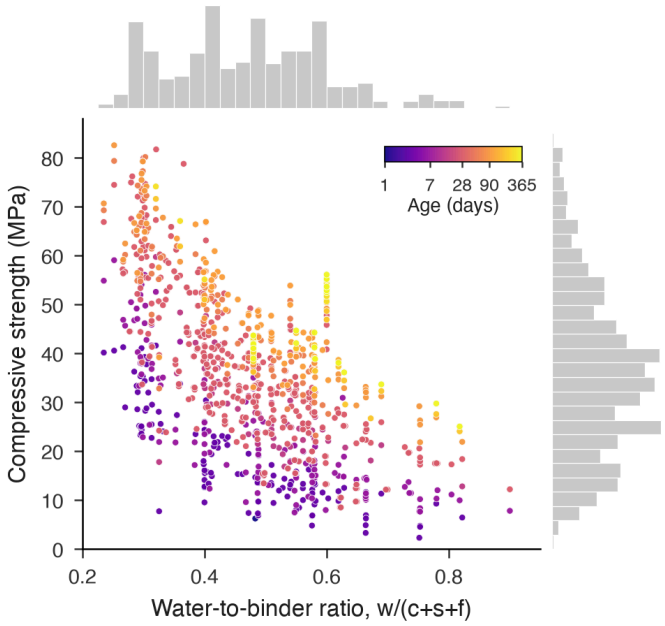


Fig. 4. Compressive strength versus water-to-binder ratio by specimen age.

Fig. 5 provides a more detailed analysis of strength development by tracking the change in strength over time for different concrete mixtures. Unlike Fig. 1, Fig. 5 examines measurement data for the same mix design at multiple ages. The strength of most mixtures increases rapidly during the early curing period, and the rate slows down later. Mixtures with lower water-to-binder ratios usually can achieve higher compressive strengths.

Fig. 5 (b) provides a more detailed comparison by normalizing the compressive strength of each mixture relative to 28-day strength. The normalized strength increases from approximately 0.46 at day 3 to 0.63 at day 7, and further to 0.82 at day 14. By approximately 90 days, the median reaches 1.25 indicating that the typical strength is about 25% higher than the corresponding 28-day value among the mixtures represented at that age. However, the shaded percentile ranges show variation in strength development across different mixtures. Therefore, during the general trend suggests increasing strength with age, the rate of development varies among mixture designs.

Based on exploratory data analysis, the compressive strength of concrete depends on multiple factors rather than any single mix design characteristic. The age of the test specimens and the cement content are positively correlated with strength, while the water content and the water-to-binder ratio are negatively correlated.

II. Proposed Analysis

The objective of this project is to investigate the relationships among concrete mixture characteristics, age, and compressive strength and to develop predictive models for concrete compressive strength. The first stage of the analysis will use exploratory data analysis to examine the variable distributions and relationships among the mixture components, age, and compressive strength. Appropriate graphical and statistical methods will be used to explore these relationships, while feature importance analysis will help identify which mixture characteristics are most strongly associated with compressive strength.

The second stage of the project will apply multiple predictive modeling approaches introduced in the course. These may include statistical regression methods and machine-learning models, as appropriate based on the methods covered during the semester. The models will be developed using the eight input variables and evaluated using appropriate prediction-performance measures. Their predictive capabilities will then be compared to determine how different statistical and machine-learning approaches perform in estimating con-

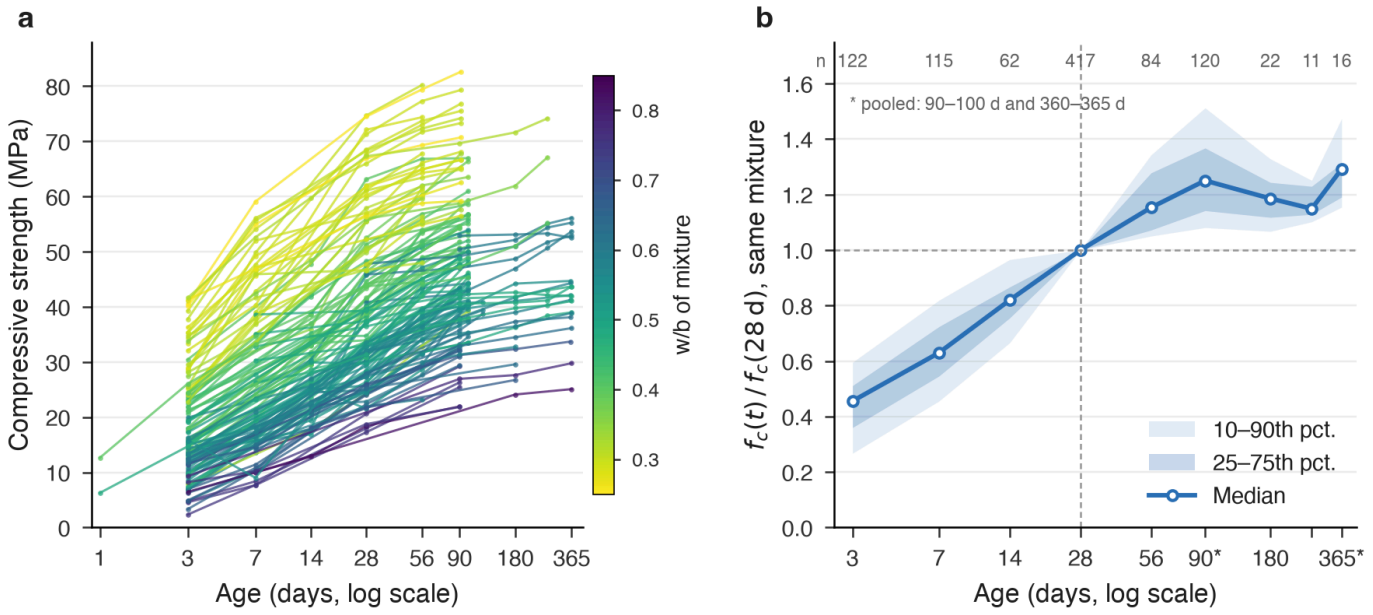


Fig. 5. Strength development of individual concrete mixtures.

crete compressive strength. This analysis is useful because compressive strength is an important indicator of concrete performance, while strength development depends on several interacting mixture characteristics and the age of the concrete. Comparing multiple predictive approaches can also provide insight into whether more flexible machine-learning methods offer improved predictive performance compared with conventional statistical regression models. Furthermore, identifying the most influential mixture characteristics can provide insights into the factors associated with concrete strength and help guide future mixture design adjustments to improve concrete performance.

III. Predictive Modeling

We will compare four models for concrete compressive strength: multiple linear regression, polynomial regression with ridge regularization, a regression tree, and random forest (RF). All models will use the seven mixture ingredients and age to predict strength in MPa. The aim is to assess whether more flexible models improve predictions for mixtures absent from the training data. The resulting models could support preliminary mixture assessment within the conditions represented by the dataset.

Multiple linear regression will provide the baseline. The polynomial model will include the eight original predictors, their squared terms, and pairwise interactions. A ridge penalty will limit coefficient size to reduce overfitting. The regression tree will divide observations through successive rules, while RF will average predictions from 500 trees built with random variation in observations and candidate predictors [2]. All models will be implemented in Julia.

The same data split will allocate approximately 80% of observations to training and 20% to testing. Records with identical quantities of the seven ingredients will stay together, including measurements at different ages. These mixture groups will be assigned intact, so the final percentages may differ slightly from 80/20. This approach will assess predictions for unseen recipes and prevent the same recipe from appearing in both sets. Repeated records will be reviewed before the split, and any corrections will apply to all models.

The three tunable models will use grid search with 20 predefined parameter configurations each. Every configuration will be assessed through the same five validation folds within the training set, with mixture groups kept intact. The proposed grids are:

- **Polynomial regression with ridge regularization:** 20 penalty values spaced evenly on a logarithmic scale from 0.0001 to 10,000.
- **Regression tree:** maximum depths of 3, 5, 8, 12, or no depth limit, combined with minimum leaf sizes of 2, 5, 10, or 20 observations.
- **RF:** maximum depths of 10 or no depth limit, minimum leaf sizes of 2 or 5 observations, and 2, 3, 4, 6, or 8 candidate predictors per split. The number of trees will remain fixed at 500.

These grids provide the same number of candidate evaluations for each tunable model, although the parameters and their ranges differ by model. Ordinary linear regression has no comparable tuning parameter and will be evaluated on the same validation folds. Data partitions and random seeds will be recorded and reused.

All models will use the same eight source predictors and untransformed strength values. Polynomial terms will be fixed before tuning. For the regression models, input columns will be standardized using means and standard deviations calculated from each training fold. The corresponding validation fold will receive the same transformation. Polynomial terms will be standardized after expansion. Trees will use unscaled inputs because their split rules do not require standardization. After tuning, scaling parameters will be recalculated from the full training set and applied unchanged to the test set.

Mean validation root mean squared error (RMSE) will determine the settings for each model and the preferred model. All four models will then be fitted to the full training set and evaluated once on the same test observations. Test RMSE, mean absolute error (MAE), and R^2 will be reported, with RMSE and MAE expressed in MPa. Predicted-versus-observed and residual plots will show where errors occur. Test results will not guide further tuning or changes to preprocessing.

SHapley Additive exPlanations (SHAP) will explain RF predictions [3], with approximate Shapley values calculated through ShapML.jl. Training observations will provide the reference data, and explanations will cover test observations after model settings are fixed. Mean absolute SHAP values will rank predictor contributions. A summary plot and dependence plots for the leading predictors will show contributions to higher or lower predicted strength relative to the reference prediction. These results explain model behavior and do not establish causal effects. Correlated ingredients may affect how contributions are shared.

The model's predictive ability will be limited by the range and distribution of the available data. Predictions for mix proportions or curing ages that are poorly represented in the dataset may be less reliable. The interpretation of the model's performance should be limited to dataset.

References

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[3] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems*, 2017, pp. 4765–4774.